

REV2 REFERENCE / REDESIGN IN PROGRESS

22 September 2026: the current print order is N1 nose only.

Use Manufacturing/Nose_Print_N1/A59_Nose_N1_mm.stl and Nose_N1_Print_Guide.pdf. Quantity one, millimetres, 100%.

The following pages retain the old Rev2 design for reference. They are not the current full-aircraft manufacturing instructions. Their original page numbers remain unchanged; this notice is an additional page.

Superseded for the new design

Landing gear and steering system; the old 42-piece print order; A59 cover screw holes, columns, spacers and shaped washers; the large nose pitot-access notch. Do not fabricate these from the old instructions.

N1 restores the original upper nose shell. It uses a proposed simple ledge/tape attachment and a probe passage drilled to actual hardware during assembly. It retains the original rear mating envelope.

The rest of the aircraft will be simplified around the plywood skeleton and foamboard. Remaining mounts, control construction, belly protection, installation details and mass/CG estimates still require that redesign. Do not use the old mass, CG or gear clearance numbers for the new configuration.

Existing cutting templates are retained until a reuse/change list is complete. The N1 release is a nose-only handoff, not a completed redesign of every file in this manual.

Checks and build acceptance

File checks passed for the specified revision. Physical results are recorded by the teams during the build.

Digital check	Recorded result
Plywood	25 profiles /37 pieces. Eight changed or added profiles reconstructed; maximum changed-profile volume difference 0.000215%. Original retained profiles keep their recorded edge-finishing requirements.
Printed parts	38 files /42 installed pieces. The 32 new/revised meshes are closed and manifold; maximum accurate CAD-volume difference 0.090%. Six retained aircraft meshes keep their previous checks.
Installed geometry	No detected overlaps for the tested new accessories/gear against the revised static model. The separate structural audit found inherited frame/foam contacts; relieve inside core as instructed. Battery positions, stops and sampled insertion/removal paths checked. Equipment envelopes have separate checks.
Foam	33 developed patterns /35 blanks retained. Prior maximum sampled strain 0.367% and seam mismatch 0.027 mm. New service/gear ports are marked after forming, not projected into replacement flat skins.
A4 plywood	Seven pages: five for 5 mm stock, one for 3 mm, one for 2.5 mm. Whole-page fit, minimum tiling, placement, calibration and paired marks checked.
Reference CG	522.3 mm aft-nose geometric quarter-MAC. Sliced-mass balance estimates span 517.2-532.8 mm aft-nose over 80 mm battery travel. Battery centre Y=-33.7 reaches the geometric reference in this estimate; no flight CG range is approved.

Physical checks still required

Measure stock and coupons. Dry-fit actual hardware, controls and covers. Test forming, joints, print strength and gear deflection. Weigh the completed aircraft and measure CG. The Electronics Team verifies loaded rails, sensors, radio range, failsafes and propulsion current/thrust/temperature.

Limits

Static fit and file export do not qualify structural loads, cooling, flutter, hand-launch performance or flight stability. The shop qualifies its bend process. An experienced fixed-wing pilot reviews the completed aircraft before flight.